

A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

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My experience and education at Iowa State have prepared me for a number of factors and issues. This includes things like designing systems, creating solutions to modern engineering problems, growing my teamwork skills, and identifying ethical dilemmas within any decisions that I make. When it comes to designing systems, I have learned how to design circuit systems, signal processing systems, and embedded systems. These are skills that will be very useful in my future career endeavors. This experience has also allowed me to develop my problem-solving skills. Every single day, there is a problem that needs to be fixed in my homework, projects, or exams. These challenges make my mind get used to them and how to navigate through them. My education has helped my teamwork skills through experiences like group projects. This can be seen in classes like CPRE2880 and EE3300 where complicated projects were assigned to groups of 2-4 students. Since these projects are so challenging, every person's perspective and ideas need to be evaluated and thought about. Within this, persuasive communication is also developed. If you are not convinced that your project or idea will work, neither will your team or your professor. Lastly, in terms of navigating ethical dilemmas, my experience has taught me that ethical decisions are made every day. Any one of a person's decisions can affect someone around them negatively or positively. This has taught me that no small decision should be taken for granted since it can affect someone close to you. This diverse education has given me the ability to act upon my ethical responsibility because I now know what is required of my future career as an engineer. I must take into account the perspective of each stakeholder in each decision that I make.

One instance that immediately comes to mind in terms of when I had to expand my knowledge beyond the classroom is when I started to go to tutoring session for Calculus 2. This was a course that I knew was infamous at Iowa State and in college in general. To confront this challenge, I put my ego aside and asked for a tutor where we met for an hour once a week. This helped me tremendously. Because of this, I earned a high grade in the class and truly understood what was happening and why.

For learning beyond the classroom, I truly believed that the engineering career fairs at Iowa State have benefited me. Attending these career fairs have strengthened my public speaking skills, my elevator pitches, and my interviewing skills. One instance that comes to mind is my interview with Texas Instruments after a career fair. I was asked all sorts of behavioral and technical questions that I was able to answer under pressure. This is a great skill that I am grateful to have acquired. Additionally, being a member of IEEE has been very beneficial because I have been able to attend speaker meetings with different companies within different industries. This includes companies like Micron, Texas Instruments, and Garmin. Events like these give me the chance to learn more about companies that I could be working for someday.

One situation where I had to harness my learning to address a new challenge was within my internship at Sandisk over the past summer. This internship had a requirement to be technically proficient in Python. Unfortunately, I had not taken any Python classes while at Iowa State. To address this challenge, I had to use my previous C++ coding skills and adapt those to learn Python on my own. This served to be beneficial because I was able to be proficient enough to where I got the job.

If I had to retrace my steps and restart my undergraduate journey, I would take more coding classes earlier on. I previously did not realize how much coding is required to work in the

engineering field, especially in tech. These coding languages include but are not limited to Python, C++, C, Verilog, SystemVerilog, VHDL, and Java. All of these coding languages can be seen on any tech company's application requirements. Not taking these classes earlier on in my career has forced me to learn them on my own or just be behind as a result. Even in the construction or biomedical industry, learning CAD software like Revit or SolidWorks are very important to be proficient in.

An engineering topic that I have recently embraced is the concept of state space models within automatic control systems. This is a concept that shows how changing any variable in a system can affect the system as a whole. At first, I literally had no idea what this was or why I was learning it. To combat this, I just raised my hand and asked the professor to explain my confusion of the topic. People are too scared to ask questions these days; I am not one of those people. After this, we were assigned homework that went over this very concept. After completing ten questions of state space models, I felt very confident about the subject. Just this past Wednesday we had a quiz on the concept, and I earned a perfect score.

One example of turning theoretical knowledge into a practical innovation was when I designed a temperature sensor within my EE3300 circuits class. This project brought concepts like diodes, current mirrors, and differential amplifiers together to form a temperature sensor that had a linear output between 0 and 50 degrees Celsius. This project elevated the quality of my work because I had to take an unknown concept, like temperature, and learn about its qualities in terms of what type of signal it produces. Additionally, a time when creativity was my ally in problem solving was when I had to study for a number of tests in one night. To solve this problem, I used AI as a tool to help me study the concepts as efficiently as possible.

A strategy that I have adopted to conquer complex material and engineering concepts has been to go over lecture slides after the lecture happens. This is especially beneficial if the professor uploads their annotated slides as well. This strategy has evolved over time. For my first two years of college, I never really did this. However, after taking EE2300 and EE2240 in the same semester, I knew something needed to change. Those two classes really kicked my butt. Starting my junior year, I looked at the lecture slides that the professor posted for every engineering class that I have took since.

As I look towards my future, I still think that my coding skills need to develop. To improve on this, I will be taking low credit classes that teach me languages like Perl and Java. There is not a limit on how many coding languages an engineer can be proficient in. Also, I think that my first full time job will develop these coding skills because jobs in tech rely on coding to produce things like testing scripts for hardware and firmware.